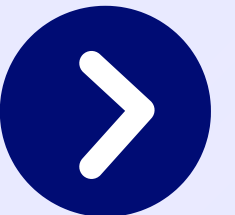


Student Pay: A Blockchain application building a wallet system for managing student spending through internal tokens.



Our team members

Nguyen Pham Quynh Trang (Leader + Main System)

La Chi Tuong (Coder)

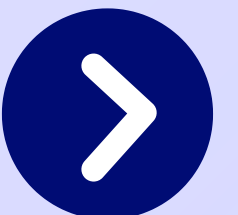
Nguyen Huu Phuc (Coder)

Do Hong Ngan (System Assistant)

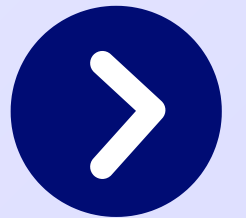


Introduction

Existing payment solutions allow parents to control spending limits but cannot control spending destinations. StudentPay is a blockchain-based programmable family wallet that enables real-time transaction monitoring, transparent payments and parent-controlled spending, helping students spend safely while maintaining financial independence.



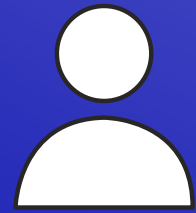
Why Blockchain?



From Soft Rules to Hard-Coded Financial Control.

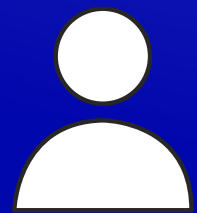
Traditional payment systems rely on application-level rules, which can be modified or bypassed and often provide limited transparency. In contrast, StudentPay leverages blockchain technology to enforce hard-coded transaction rules, automatically reject unauthorized payments, and maintain an immutable transaction history. As a result, all transactions become transparent, secure, and fully auditable.

System Roles



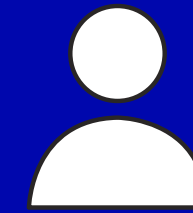
PARENT (CONTROLLER)

- Manage spending permissions
- Approve pending transactions



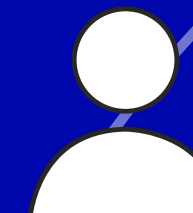
MERCHANT

- Receive payments
- Withdraw funds



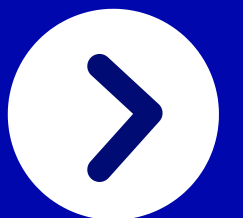
STUDENT (USER)

- Make purchases using StudentCoin

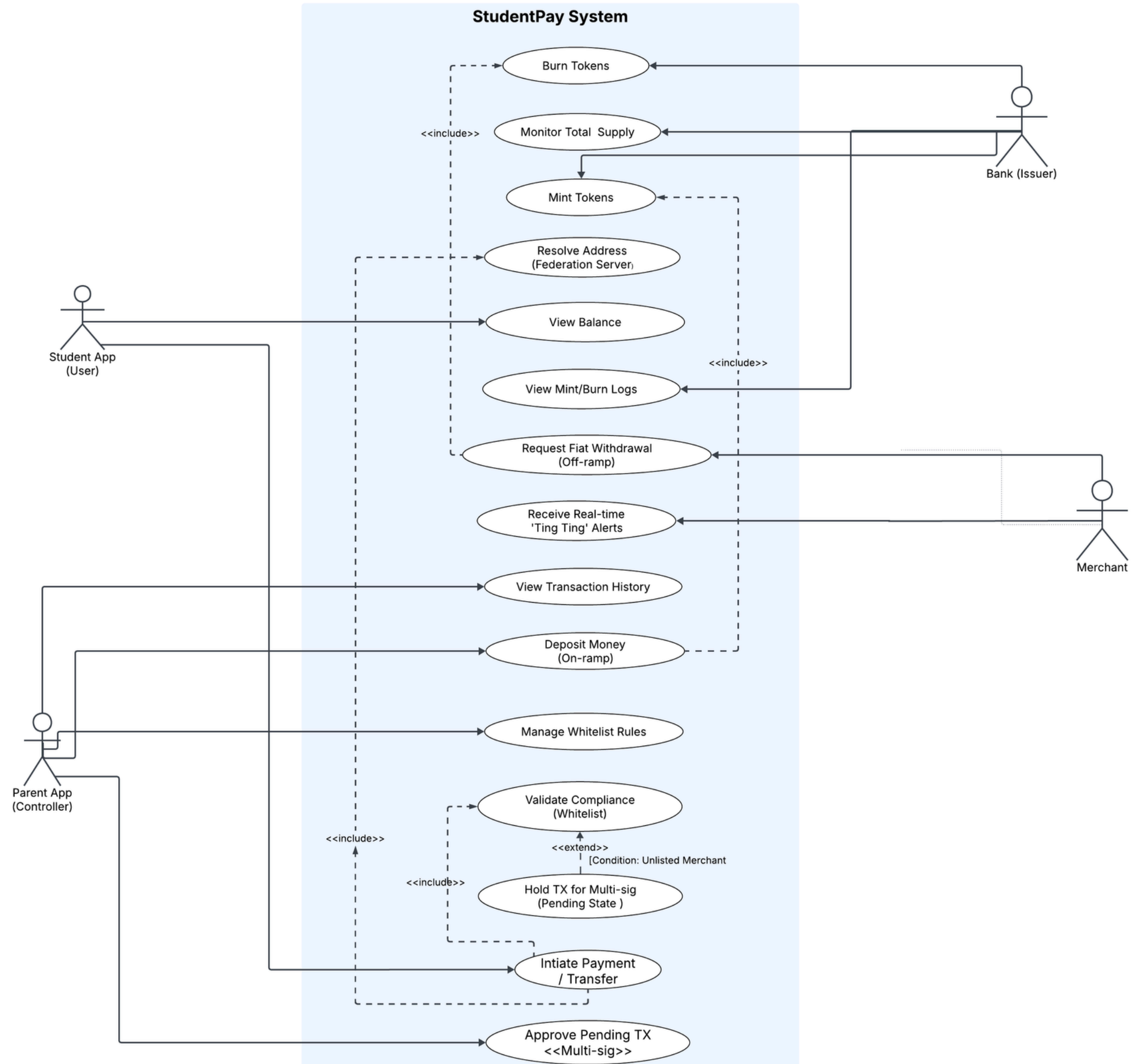


BANK (ISSUER)

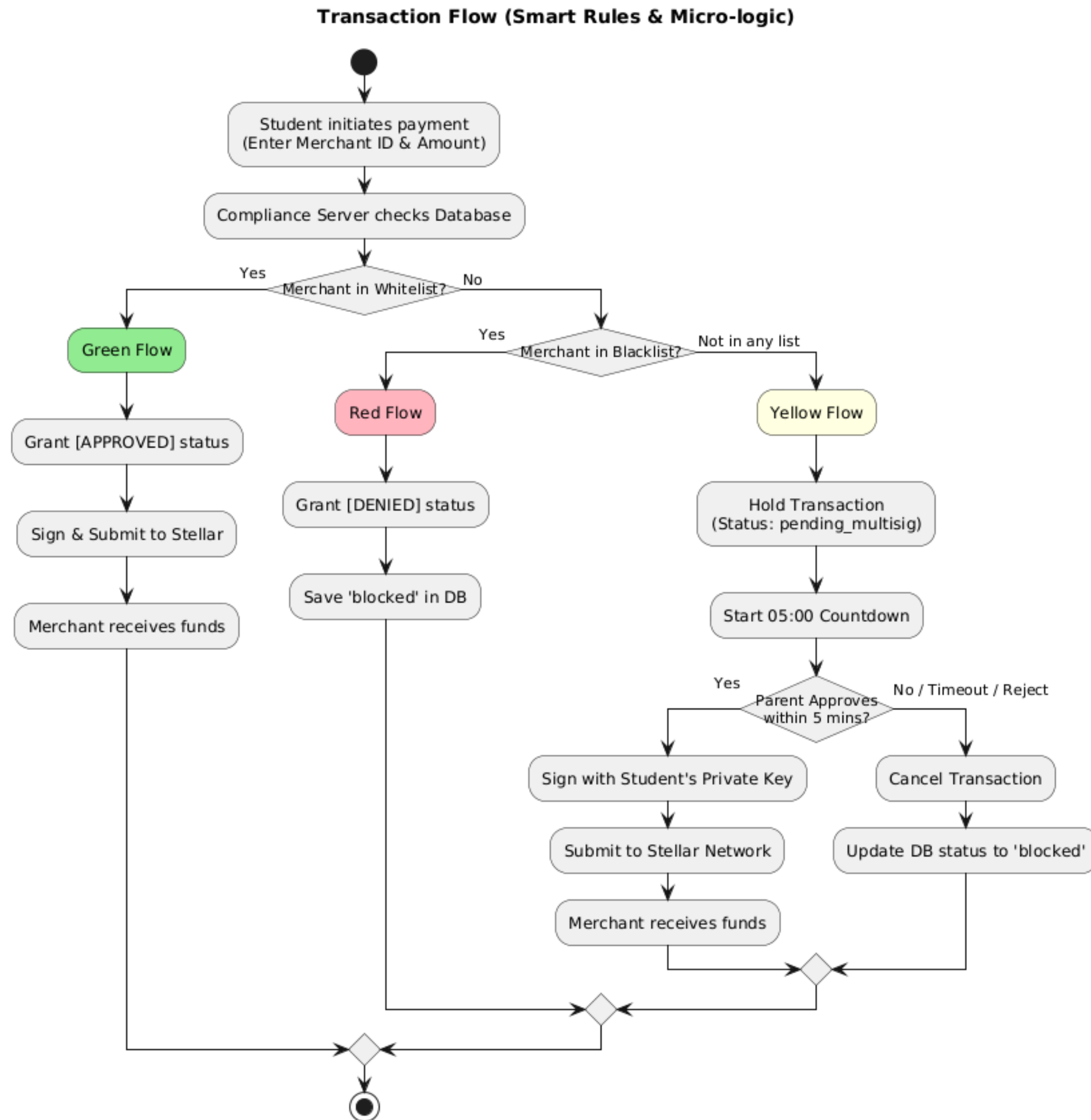
- Mint and burn StudentCoin
 - Maintain 1:1 VNĐ backing
- Design Strategy



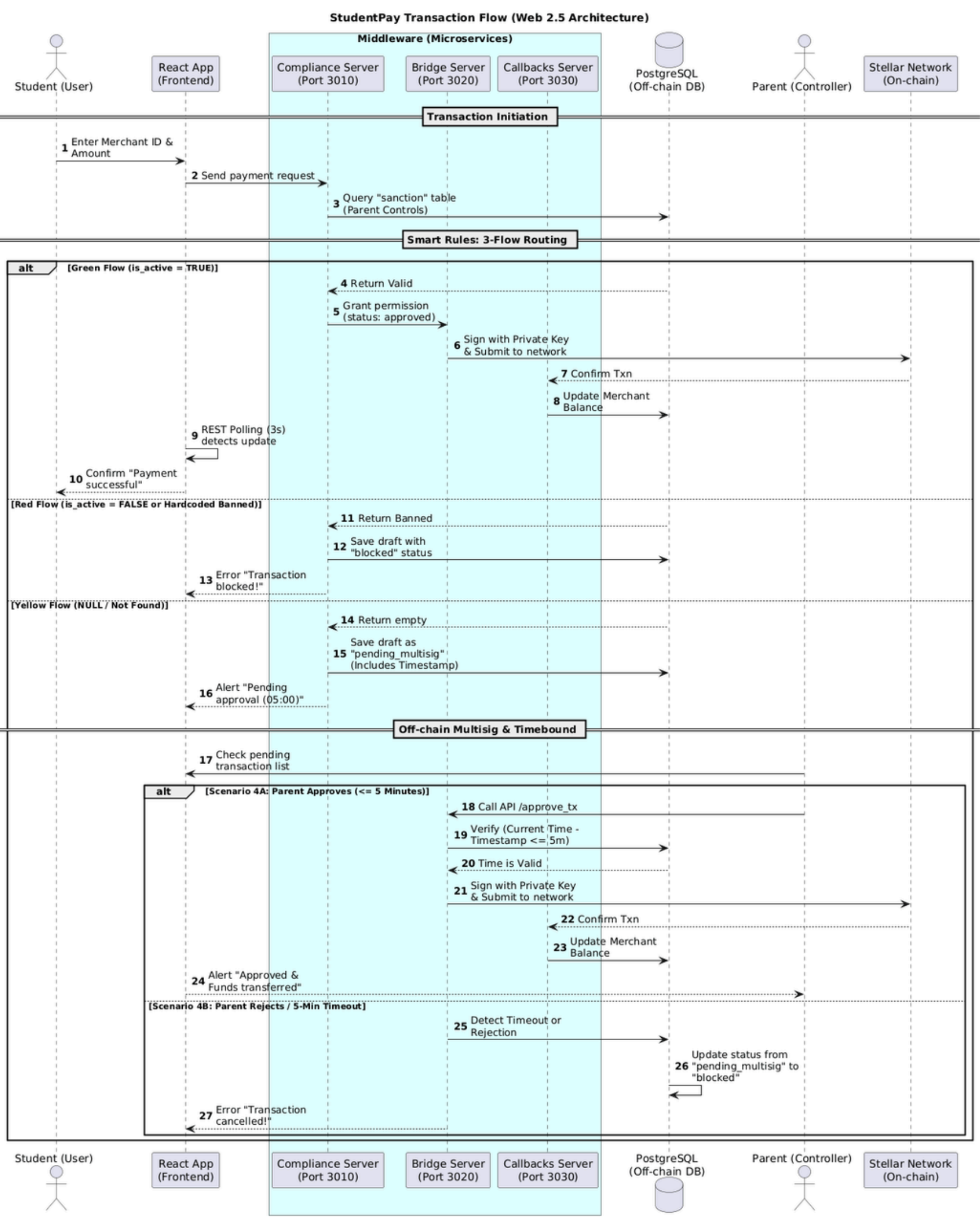
USE CASE DIAGRAM



TRANSACTION FLOW



SEQUENCE DIAGRAM



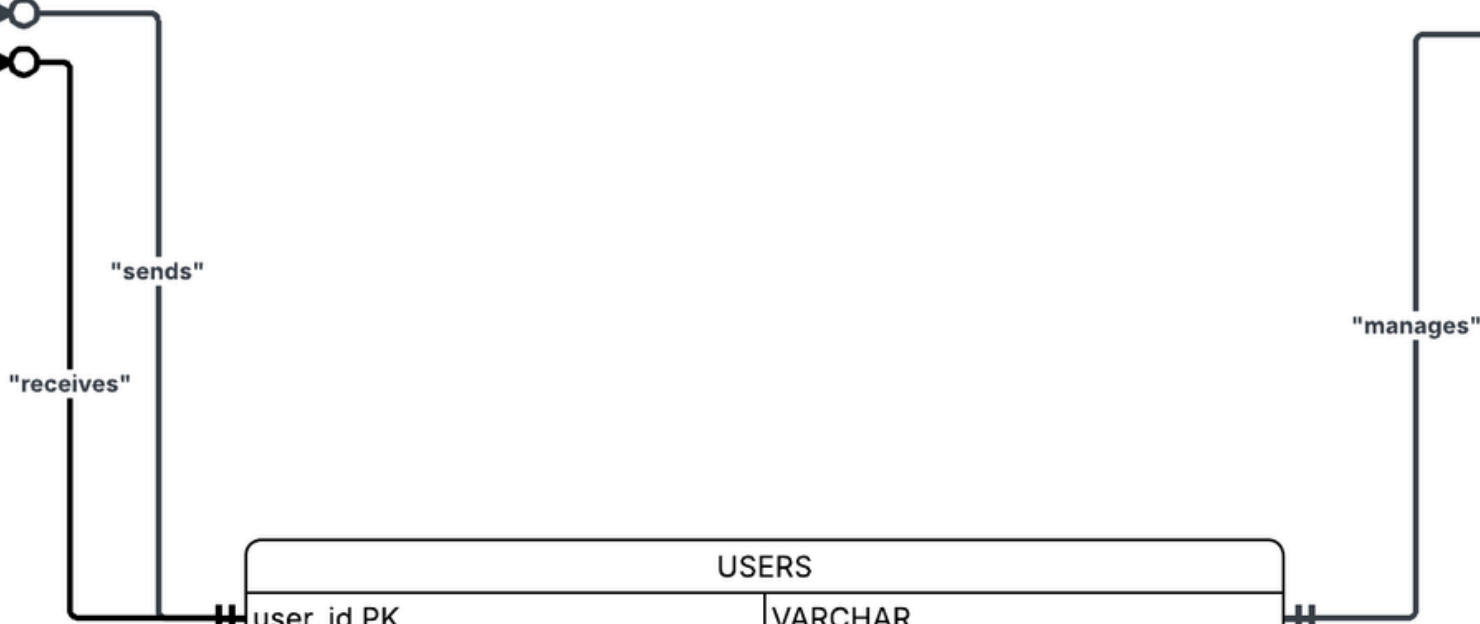
ERD

StudentPay ERD (Crow's Foot)

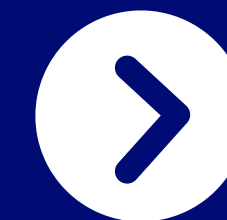
TRANSACTIONS	
internal_id PK	SERIAL
stellar_tx_hash	VARCHAR
sender_id FK	VARCHAR
receiver_id FK	VARCHAR
amount	DECIMAL
status CHECK ('approved','blocked','pending_multisig')	VARCHAR
timestamp	TIMESTAMP

USERS	
user_id PK	VARCHAR
role CHECK ('student','parent','merchant','bank')	VARCHAR
domain	VARCHAR
stellar_public_key UNIQUE	VARCHAR
balance	DECIMAL
created_at	TIMESTAMP

SANCTION	
rule_id PK	SERIAL
parent_id FK	VARCHAR
domain	VARCHAR
bankname	VARCHAR
is_active	BOOLEAN



System Architecture Technologies



TECHNOLOGIES USED

- React.js
- Node.js
- PostgreSQL
- Stellar Testnet

Transaction Processing Flow

Frontend → API → Compliance →
Bridge → Stellar Blockchain

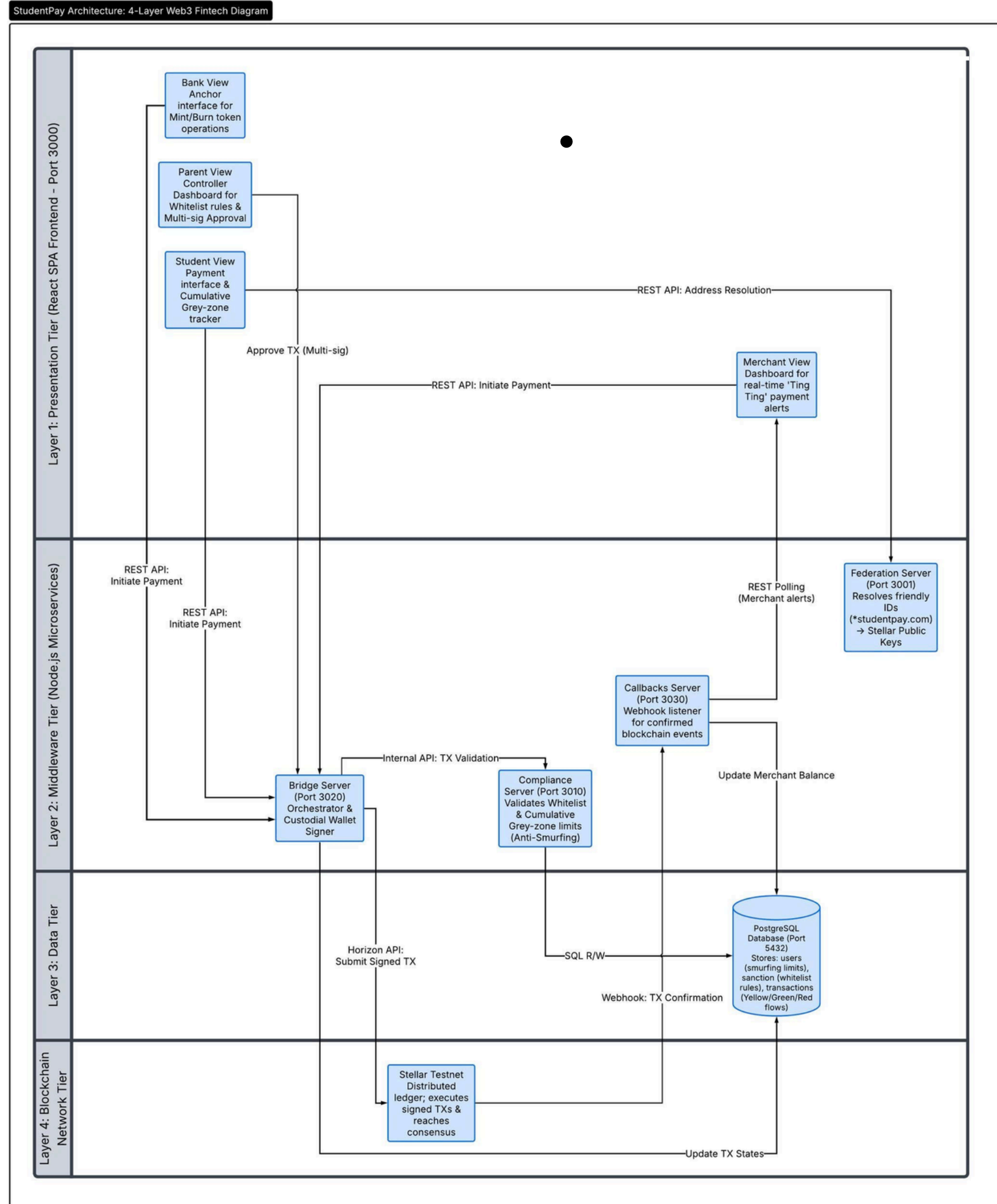
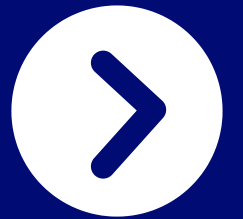
FOUR-LAYER ARCHITECTURE

- Presentation Layer: Student View, Parent View, Merchant View
- Middleware Layer: Federation Server, Compliance Server, Bridge Server, Callback Server
- Data Layer: PostgreSQL Database
- Blockchain Layer: Stellar Network

Thynk Unlimited

System Architecture

OTHERS



DEMO FRONTEND